Exercise 2: Data Exploration — Exploring Three Public APIs

For this exercise, I explored three public APIs (OpenWeatherMap, NASA Open APIs, and the Spotify Web API) to understand their accessibility, limitations, and usefulness. The OpenWeatherMap API provides weather and climate data freely through a registered key, though it limits request frequency for free users. NASA's APIs are almost completely open, offering access to scientific datasets like images of planet Earth and Mars rover photos with only minimal usage restrictions. In contrast, the Spotify API requires authentication and restricts access to user data and audio content, focusing instead on track and artist data. While OpenWeatherMap and NASA encourage educational and research use, Spotify imposes stricter controls to protect user privacy and intellectual property.

Overall, the data from these APIs is reliable, documented, and highly useful for projects in various domains from environmental and scientific research to cultural and analytics related to behaviour. NASA's openness promotes scientific learning, OpenWeatherMap supports forecasting applications in the real world, and Spotify enables analysis of musical trends. Despite some limitations on data volume and commercial use, all three APIs demonstrate how public data access can drive creativity, innovation, and meaningful analysis in data driven projects.